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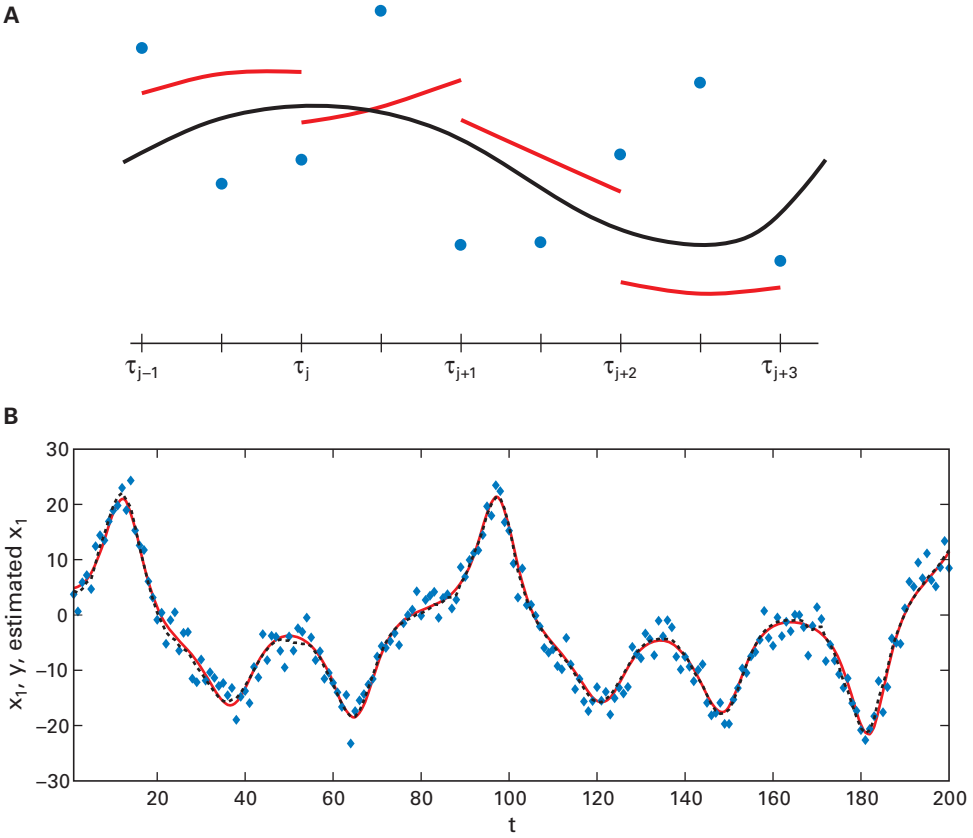


Plate 1
Multiple shooting schematic for solving inverse parameter estimation shown in A, and an actual example of noisy Lorenz system is shown in B. Reproduced from [VTK04] with permission.

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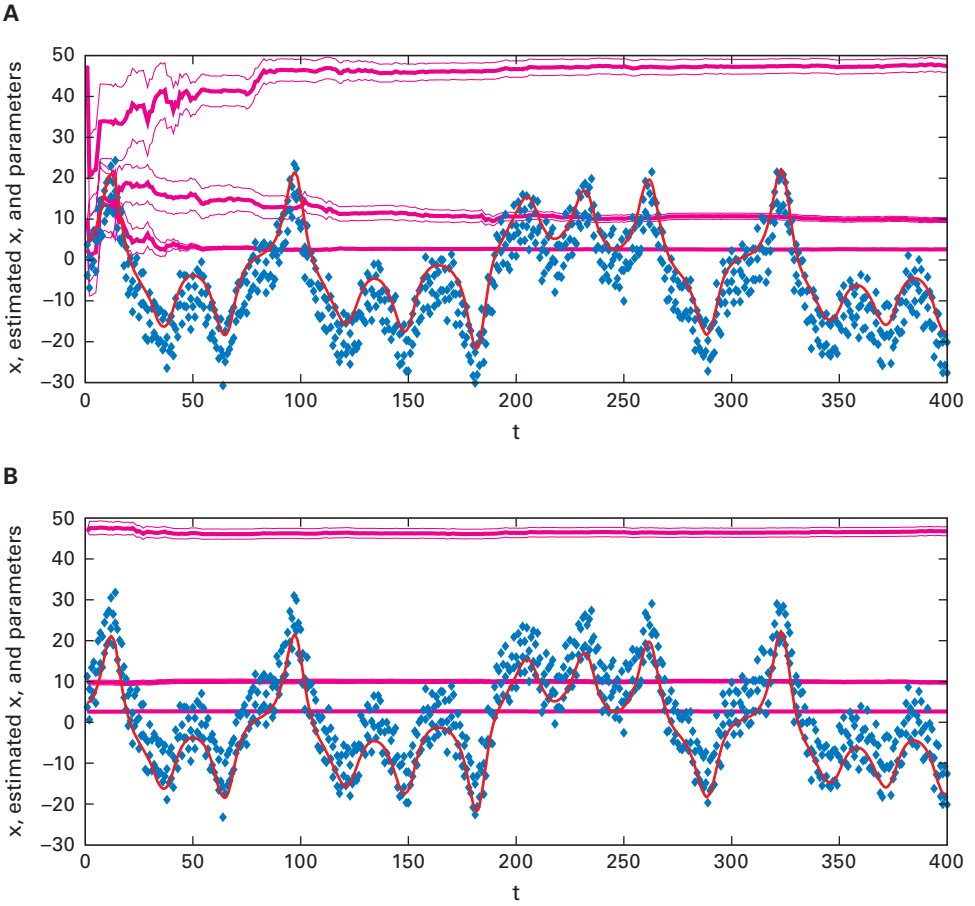


Plate 2
A shows the first pass of the unscented Kalman filter through the noisy x output (dots) of the Lorenz system. The three parameters being estimated are shown as solid lines, with thin line confidence limits gradually converging from left to right. The values at the right are then used to initiate another pass at parameter estimation, shown in the lower panel, B. Although parameter estimation improves a bit more, the upper panel demonstrates the potential power of this technique to work with one pass in real-time applications. Modified from [VTK04] with permission.

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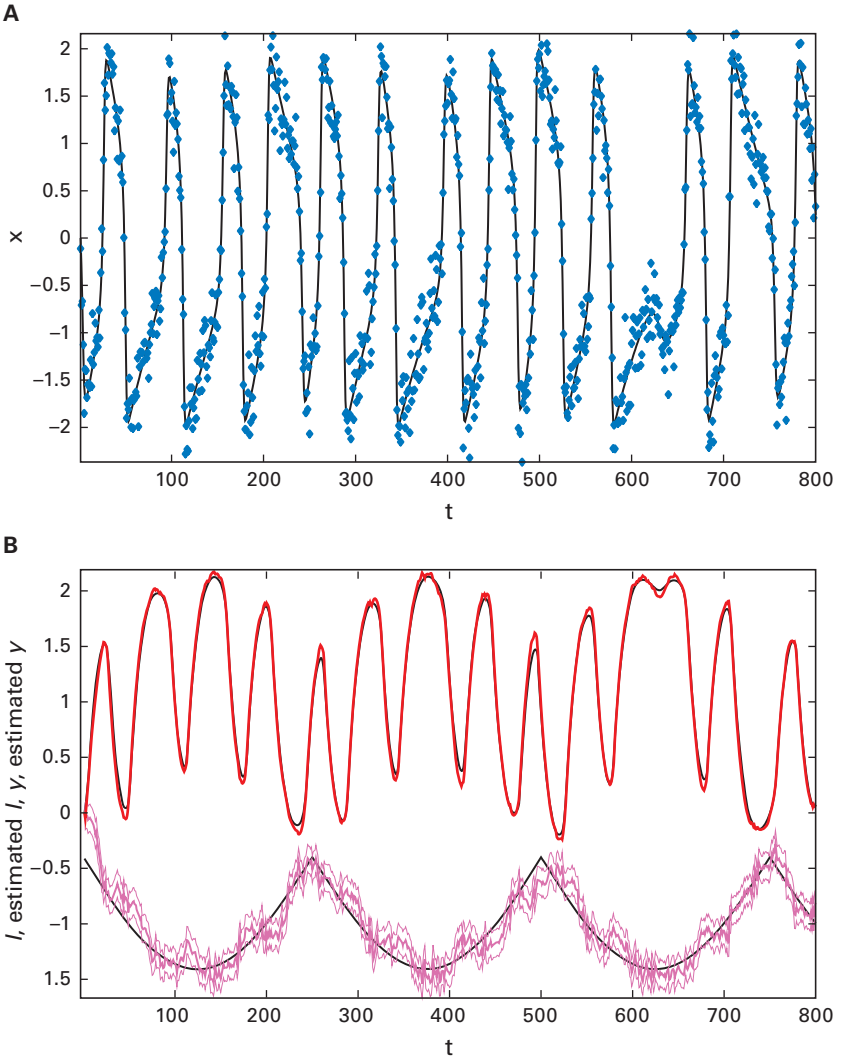


Plate 3
Variable and parameter tracking in the Fitzhugh-Nagumo equations. A shows the underlying trajectory (solid line), the noisy measurements (dots), and B shows the reconstruction (red) of the unmeasured variable y (black, nearly overlapping) in the upper half of the panel, and the applied current I(t) and its estimation as a tracked parameter (magenta) with 1 standard deviation confidence limits (thin magenta lines). Reproduced from [VTK04] with permission.

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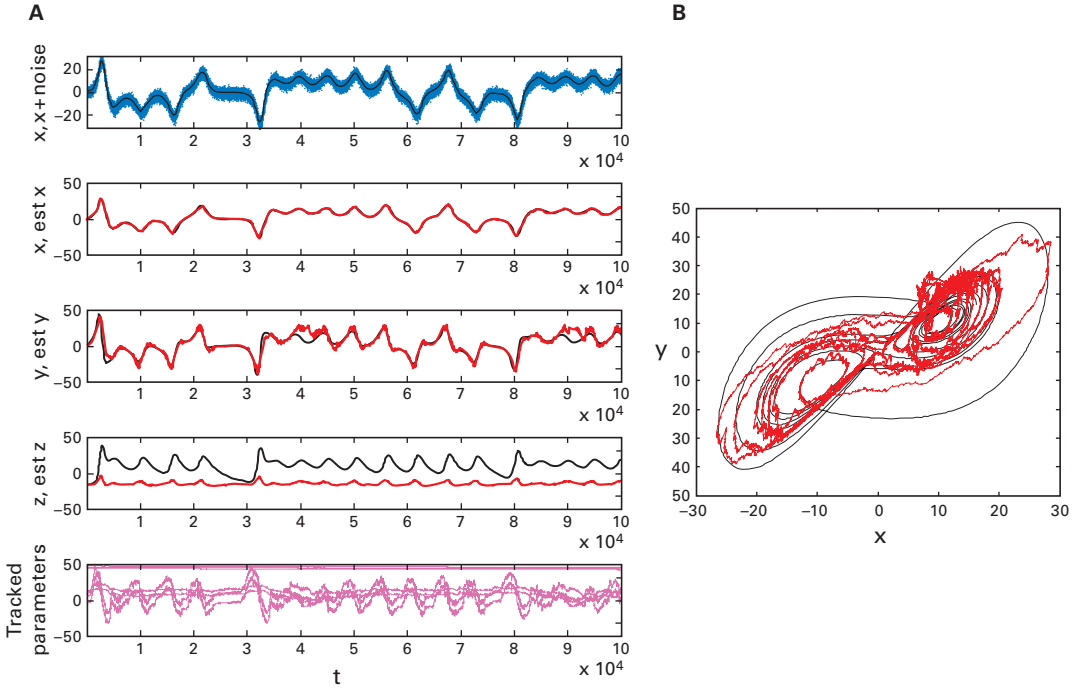


Plate 4

Results of reconstructing the Lorenz equations using the unscented Kalman filter from measuring only noisy x variable and estimating variables and parameters. In this simulation, the integration time step is 0.0001, $\mu = 10$, $r = 46$, and $b = 8/3$. A shows plots of reconstructed Lorenz variables. Upper panel is true (black) and noisy x variable (blue), second panel is true (black) and reconstructed x variable (red), third panel is true (black) and reconstructed y variable (red), fourth panel is true (black) and reconstructed z variable (red), and lower panel shows three reconstructed parameters with confidence limits. B shows true (black) and reconstructed (red) Lorenz x versus y trajectories. Note the difficulties with the z variable—similar troubles are seen in synchronizing Lorenz equations through this variable [PC90].

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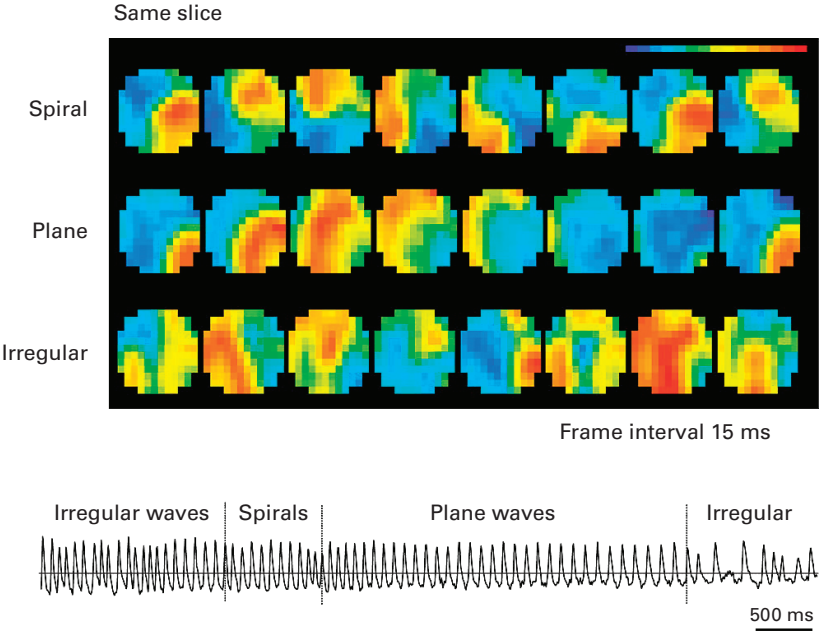


Plate 5
From a single oscillatory episode, a progression is seen of chaotic irregular, to spiral, to plane, and back to chaotic waves. Reproduced from [HTY+04] with permission.

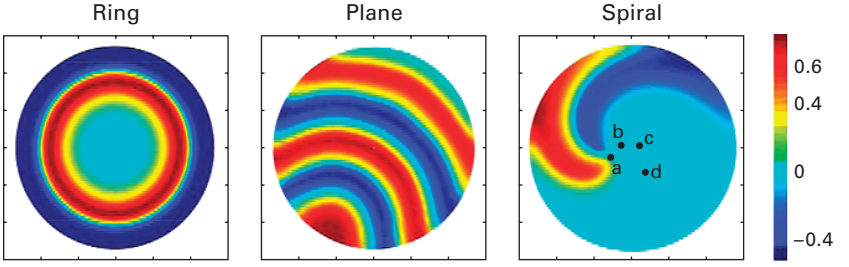


Plate 6
Simulation of ring, plane, and spiral waves with the Wilson-Cowan equations. Reproduced from [HTY+04] with permission.

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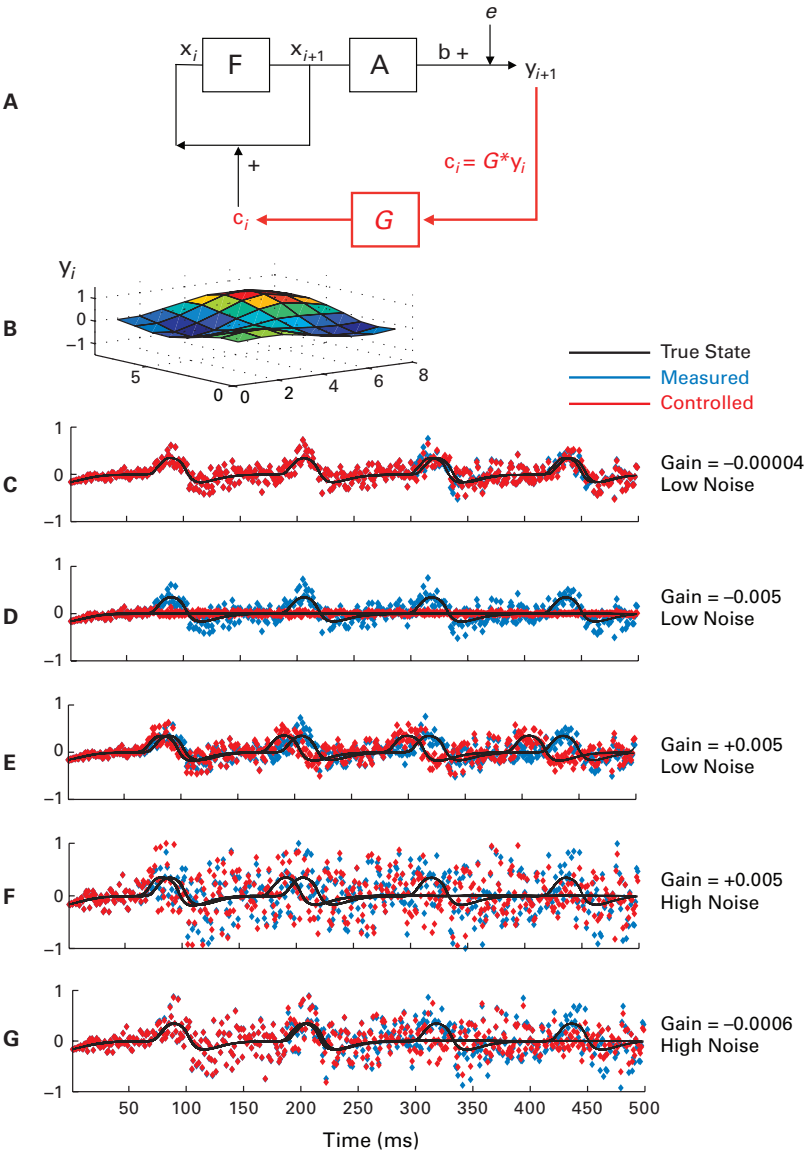


Plate 7

(A) Schematic of dynamical system *F*, which acts on state variable vector *x*, and observable function *A*, which produces output vector *b*, to which measurement noise *e* is added, generating output observable *y*. When direct proportional control is used, control function *G* is employed, which generates the control vector *c*. This control vector is directly "injected" into the neuronal elements, adding vector *c* to the activity variable *u* (nothing is added to the unobserved variable *a* or to the parameters of the system). (B) An instantaneous snapshot of the 8 × 8 output *y* during a rotational spiral. (C) The output of the observable *y* under conditions of small negative gain and low noise, from a single element of the grid in B, shown as a time series without (blue) and with (red) control applied. The underlying state vector activity element u_i is plotted as a thin black line. In contrast with C, D shows outputs under conditions of higher negative gain and low noise, E shows positive gain and low noise, and F and G show high noise conditions with positive and negative gains, respectively. Low noise corresponds to 1 to 5, and high noise corresponds to 10 to 14 on an arbitrary scale. Note that under high noise conditions, the application of direct proportional control as illustrated in the schematic renders the system dynamics unstable and destroys the rotational spiral regardless of positive or negative feedback. Reproduced from [SS08] with permission.

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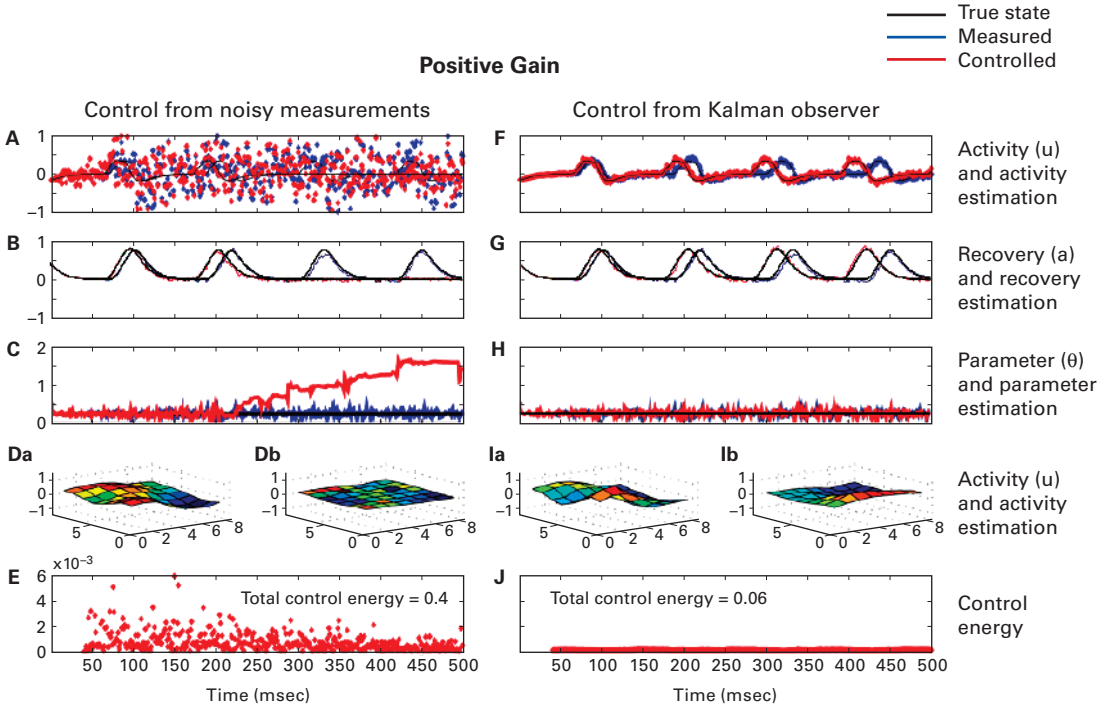


Plate 8

Comparison of the use of positive gain proportional control calculated directly from high noise observations, left column A–E, and from the Kalman observer, right column F–J. (A) Output variable y_i from single element of grid, is the same as in figure 6.21F. The application of positive feedback gain at this high level of noise results in instability of the rotational oscillation, with loss of both the underlying oscillation u_i (black line) and measured observation y_i (red). (B) Without control being applied, the Kalman observer can accurately estimate (blue) the unobserved variable a (black). With this level of noise, when control is applied, not only is the observable a oscillation lost (red), but the true system state loses stability and the wave is destroyed after two oscillations (black). (C) Without control, despite the high noise level, the parameter θ (black) is accurately tracked (blue). When the stability of the dynamics is lost during control, the parameter θ is no longer accurately estimated (red). Da shows the variable y, and Db the estimated underlying activity $\hat{u}$. The snapshots in Da and Db are taken from the last frame in the 500 msec simulation, and the full simulation can be viewed as Supplementary Movie 1 from [SS08]. In E is shown the energy (sum of squared voltage) of the control vector c_i, and the total sum of all control vector energy for the 500 msec is 0.4 (arbitrary scale). F shows the estimated observable $\hat{y}$ without (blue) and with (red) control derived from the Kalman observer. (G) With Kalman observer control, the unobserved recovery variable a is now well tracked and the rotational spiral is stably increased in frequency, and in H, the parameter θ is now well tracked. Ia and Ib show the experimental variable y and estimated underlying variable $\hat{u}$. In J is shown that the control energy required is markedly reduced, with total control energy 0.06 (arbitrary scale).

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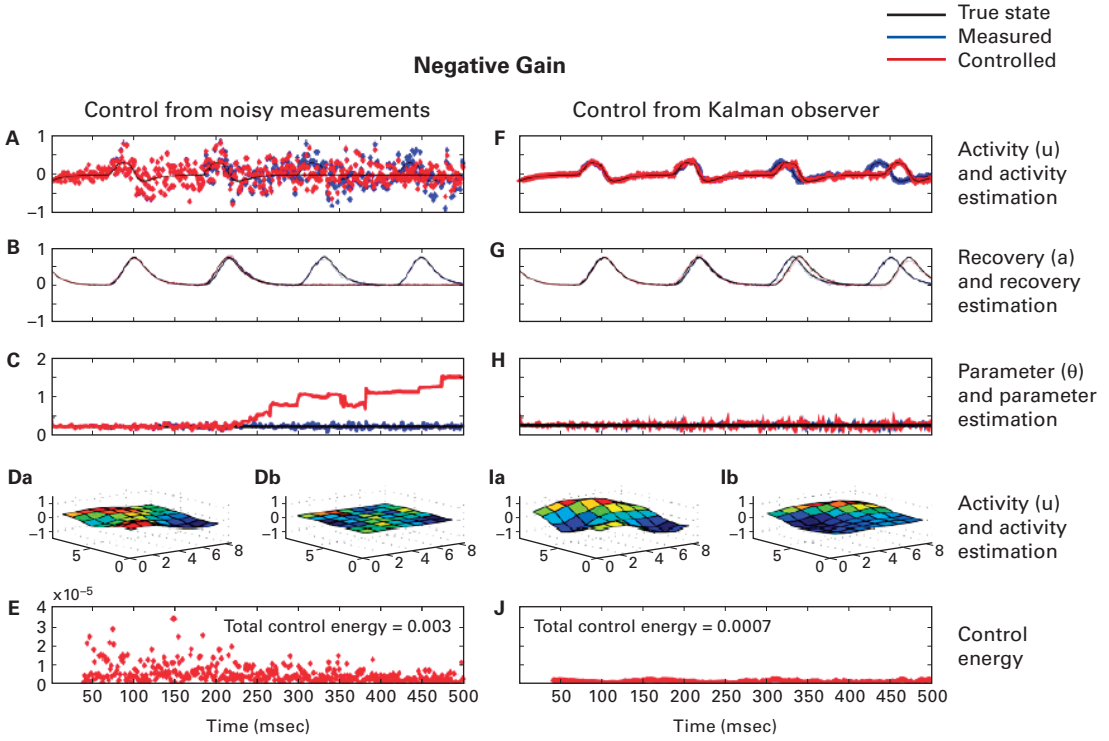


Plate 9

Comparison of the use of negative gain proportional control calculated directly from high noise observations, in left column A–E, and from the Kalman observer, in right column F–J. (A) Output variable y_i from single element of grid, same as in figure 6.21G. The variables in this figure are similar to the description for figure 6.23. Note that with control derived from the Kalman observer (right column), negative gain in high noise conditions stably slows the frequency of the spiral, and requires considerably less control energy than the unstable case where direct proportional control is calculated from the noisy observations (left column).

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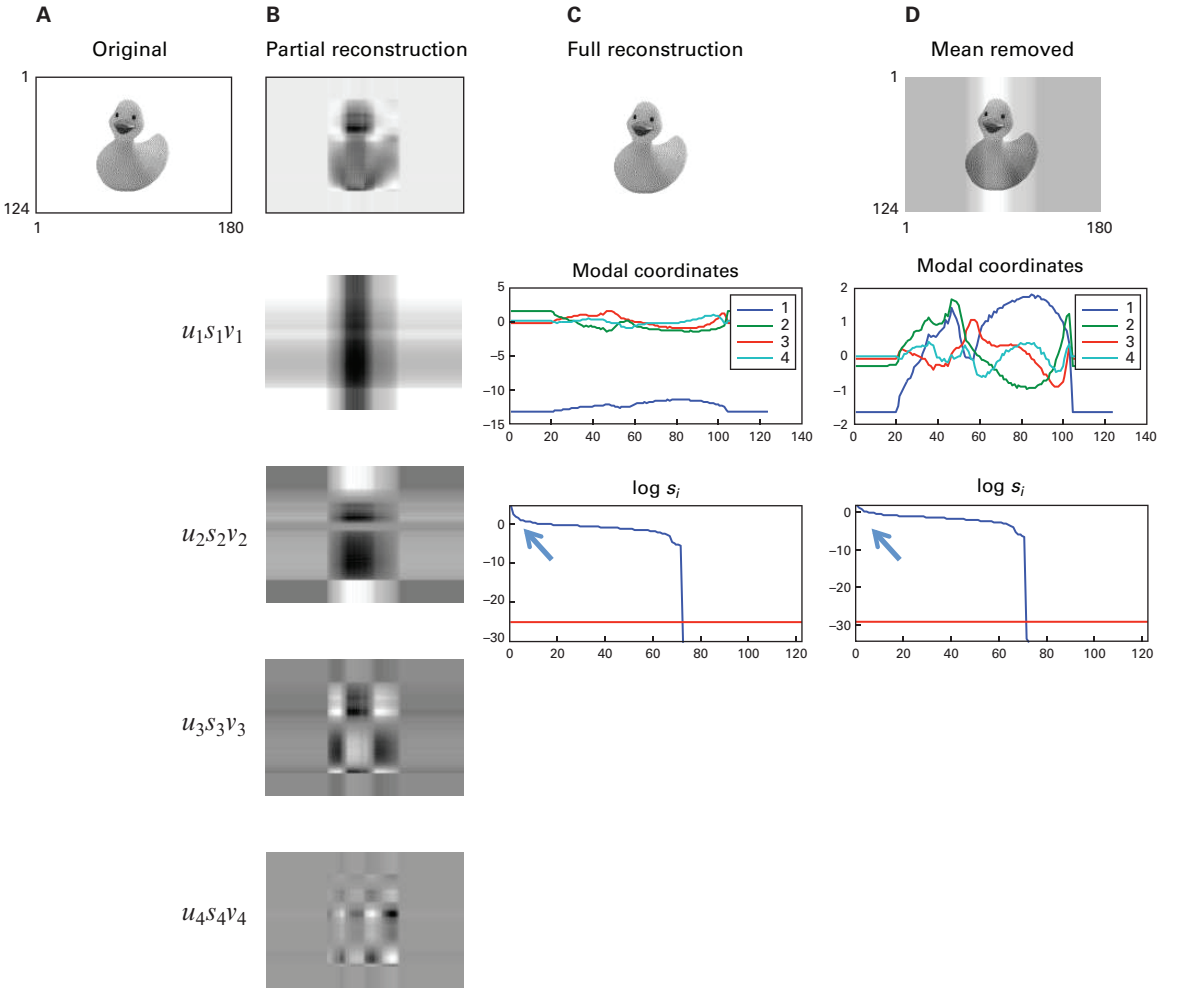


Plate 10
Singular value decomposition of a popular image in A. In B are four matrices $u_i\ s_i\ v_i$ and the sum of these matrices is shown at the top. C shows a full reconstruction with the mean retained in the image. D shows the image with the mean removed. The arrows point to the difference that the large first mode contributes to the spectrum of singular values s_i. The horizontal red lines indicate a tolerance (red lines) based on the product of the largest of the singular values times the computer machine precision—anything below the tolerance line is computationally meaningless.

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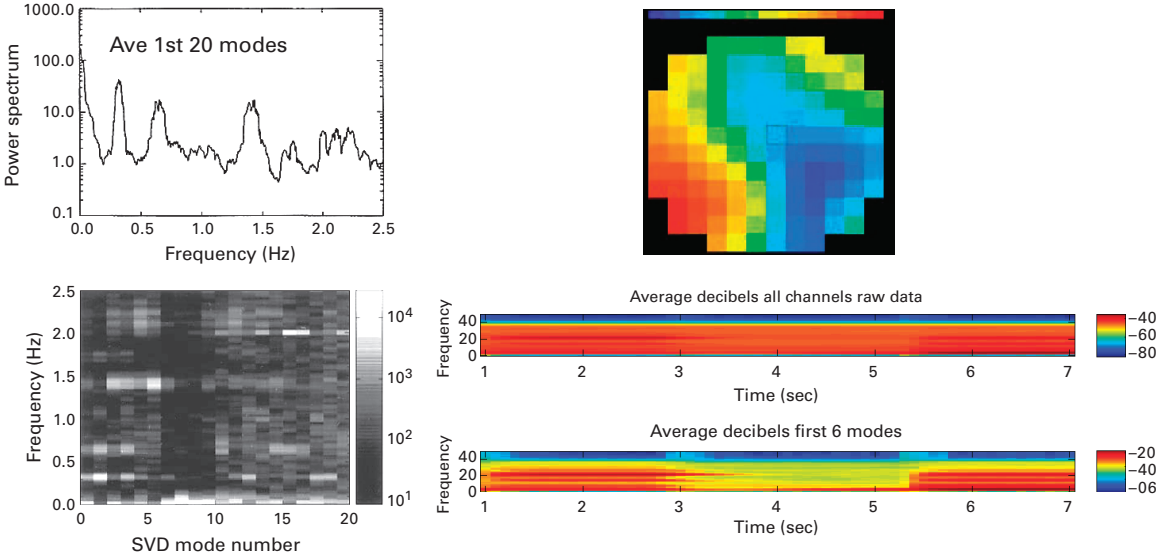


Plate 11
On left is an example of modal filtering with multitapered average spectrum from fMRI dataset, and below that, the spectrogram from individual modes. Reproduced from [MP99] with permission. On the right is an example of our spiral cortical images, with unfiltered and modal-filtered (retention of first six modes) spectrograms. The modal filtered version shows the dynamical evolution much more clearly.

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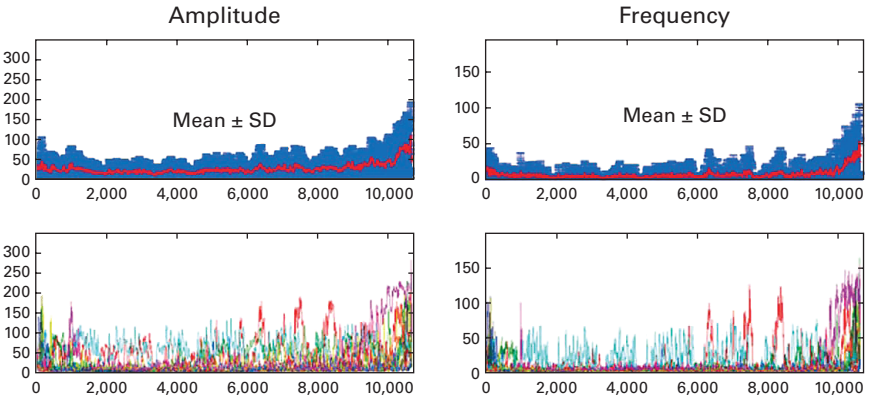


Plate 12
Effective instantaneous dimension shown for each resampled data set, below, and as mean and standard deviation, above, for amplitude and for spatial frequency. Reproduced from [SHW07] with permission.

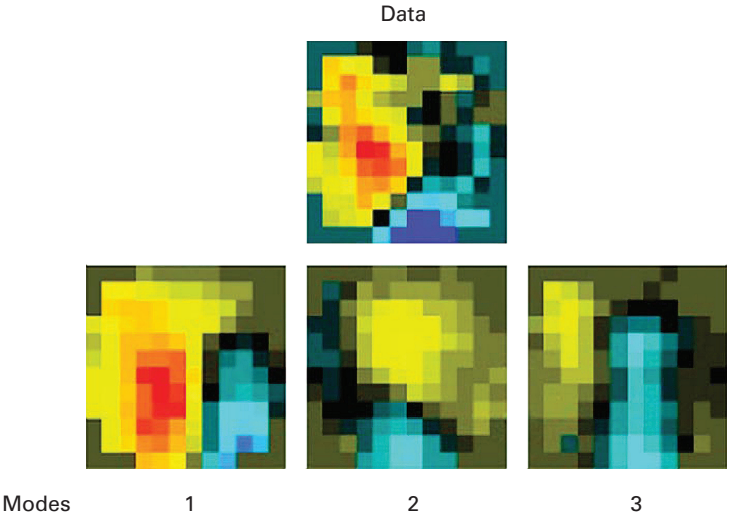


Plate 13
Mode decomposition of a typical snapshot of spiraling cortical dynamics. Reproduced from [SHW07] with permission.

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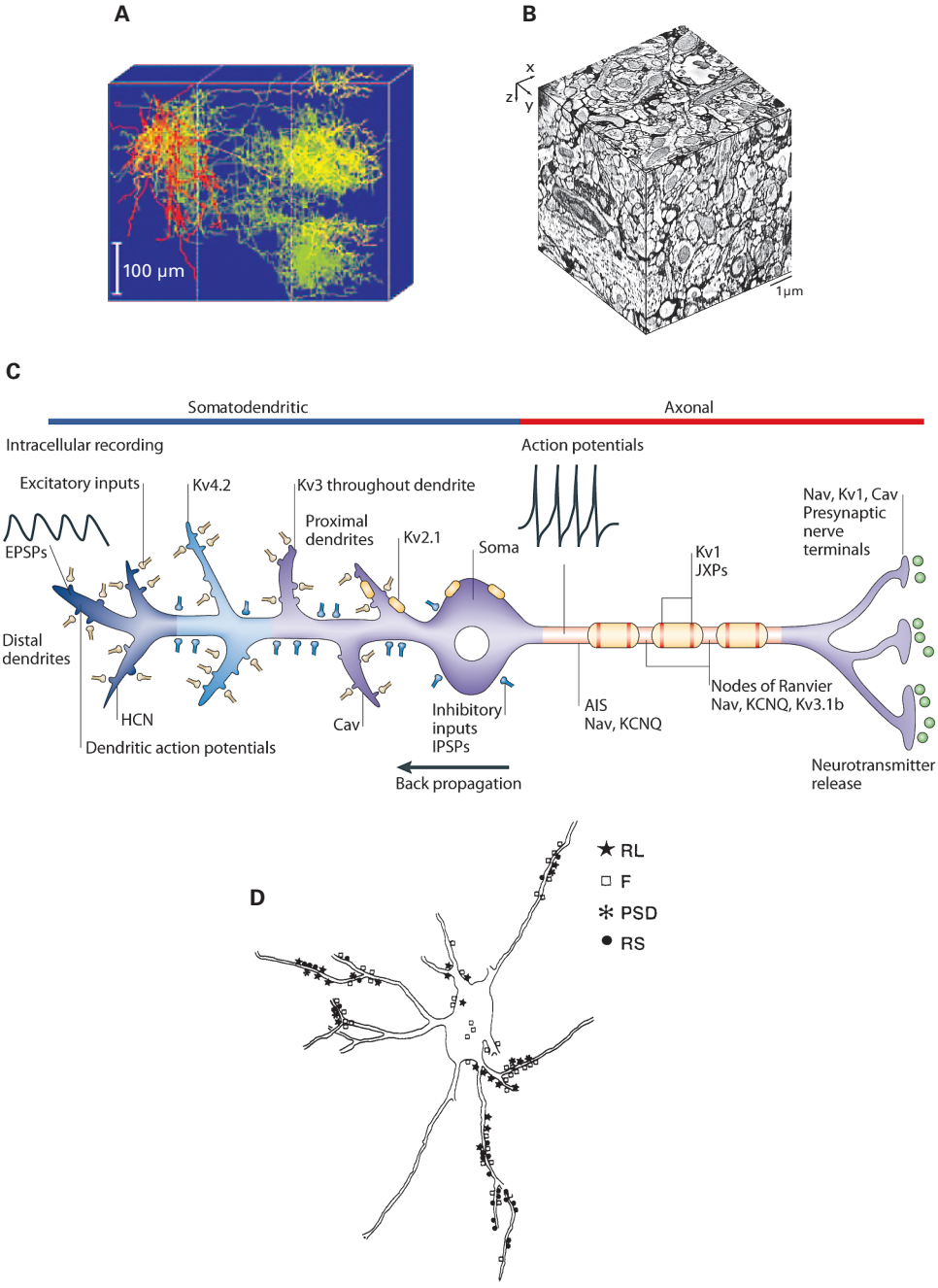


Plate 14
Complexity of neurons and their networks. A shows the reconstruction of six individual interneurons in the hippocampus, whose dendrites are shown in red, and axons in yellow. Reproduced from [GDR+05] with permission. B shows an electron micrograph reconstruction of a 350-μm reconstruction of a small cube of mammalian neuropil into which one must envision the neurons from A are embedded. Reproduced from [BD06] with permission. C shows an inhomogenous distribution of voltage-gated ion channels in a neuron. Reproduced from [LJ06] with permission. D shows an inhomogenous distribution of synapse type on a neuron. Reproduced from [LHJ95] with permission.

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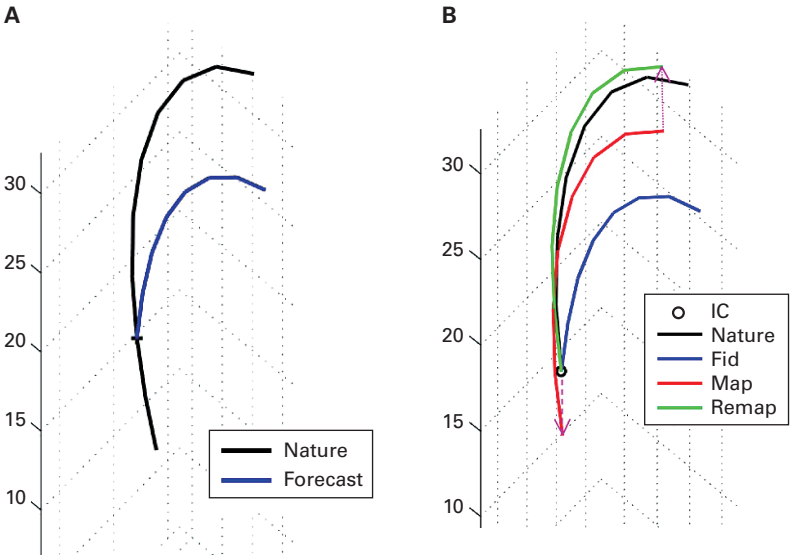


Plate 15
A illustrates imperfect model trajectory. B shows mapping strategy to bias the initial condition, IC, in order to improve the trajectory. Fid indicates the fidelity paradigm. The bias applied to the initial condition in mapping (Map) is corrected following iteration (Remap). Reproduced from [TP07] with permission.

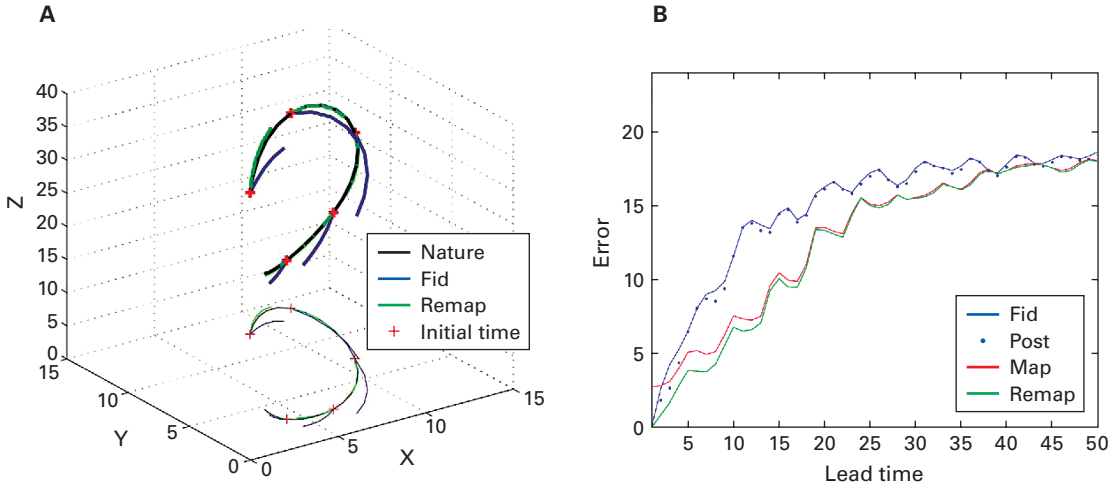


Plate 16
Sequential application, A, of the mapping-remaping (green) versus the fidelity (Fid) paradigm in the two forms of the Lorenz equations as described in the text. On the right are shown the sequential errors between the two methods. Standard post-processing (Post) to remove bias. Reproduced from [TP07] with permission.

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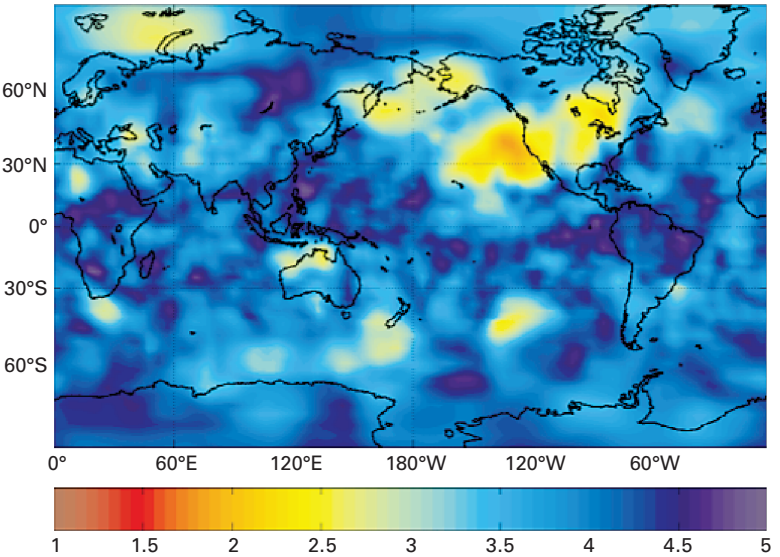


Plate 17
Bred vector dimension for wind speed across the world. Reproduced from [PHK+01] with permission.

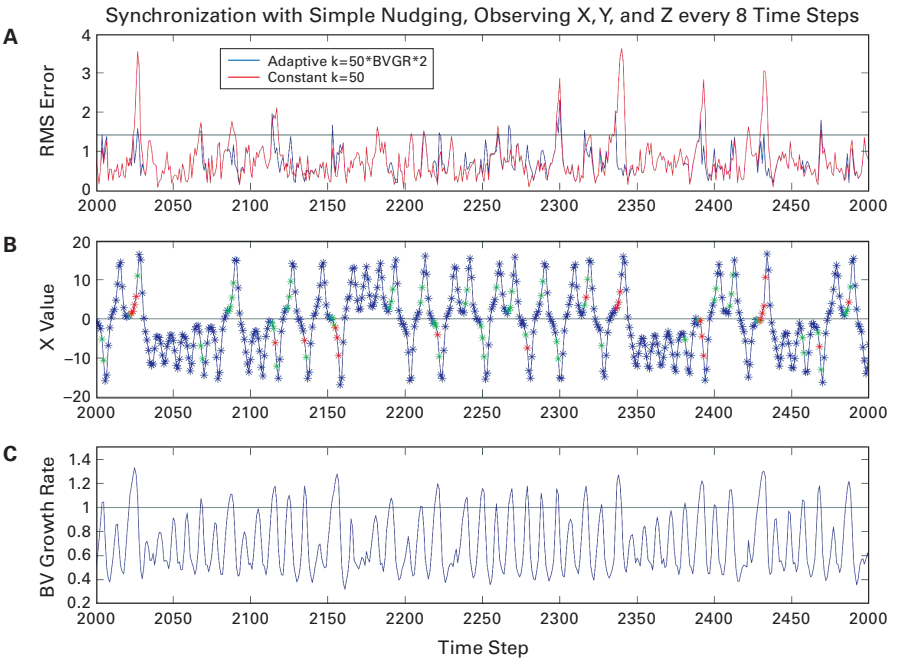


Plate 18
Nudging synchrony control for Lorenz 1963 equations using simple nudging along the axes of the original variables (blue), versus along the bred vector directions (red), shown in panel A. Panel B shows the Lorenz x in the slave system, along with the color-coded value of the magnitude of the singular value for the bred vector with blue being small, green being medium, and red being large in amplitude. Panel C shows the logarithmic bred vector growth rate, each averaged over 8 time steps. This growth rate is predictive of impending desynchronization. Reproduced from [YBL+06] with permission.

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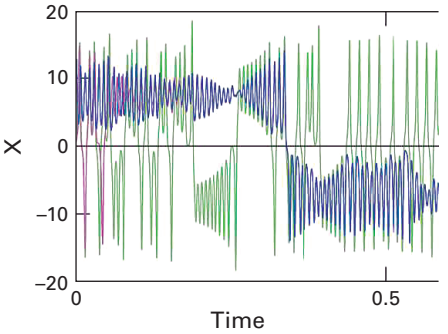


Plate 19
Two identical auxiliary slave systems, in pink and blue, driven by a dissimilar drive system in green. The auxiliary systems synchronize rapidly to each other. Modified from [YBL+06] with permission.

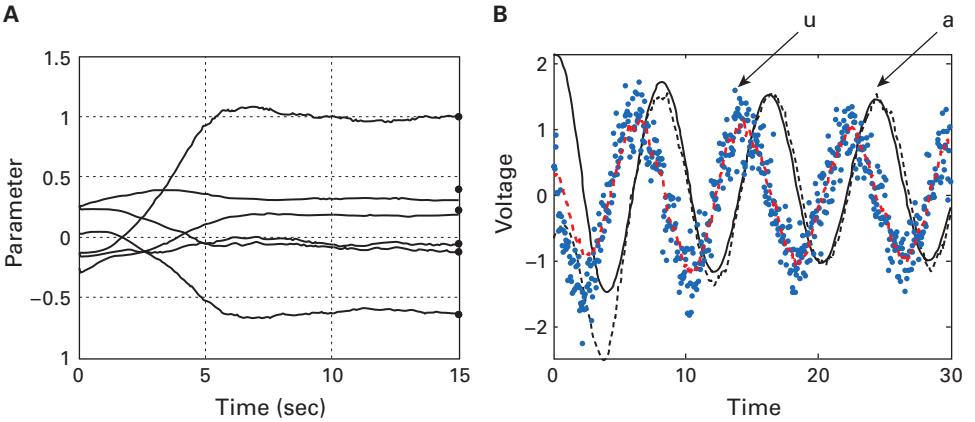


Plate 20
A shows that when tracking a heterogeneous network, the ensemble Kalman filter can be used to let the parameters float and find average values that converge close to the actual mean parameter values in the network being tracked (filled circles). B shows that, from one of the grid points, the u and a variables are closely tracked. Reproduced from [SS09] with permission.

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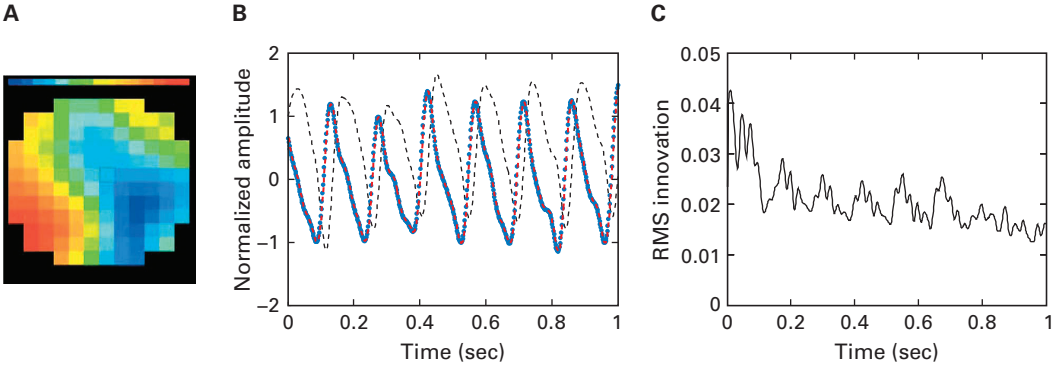


Plate 21
A snapshot of voltage-sensitive dye data from a sequence of thousands of snapshots collected at 1,000 samples per second (A), the consensus set tracking of a single Wilson-Cowan node (B), and the root mean square (RMS) convergence of the tracking errors (C). Reproduced from [SS09] with permission.

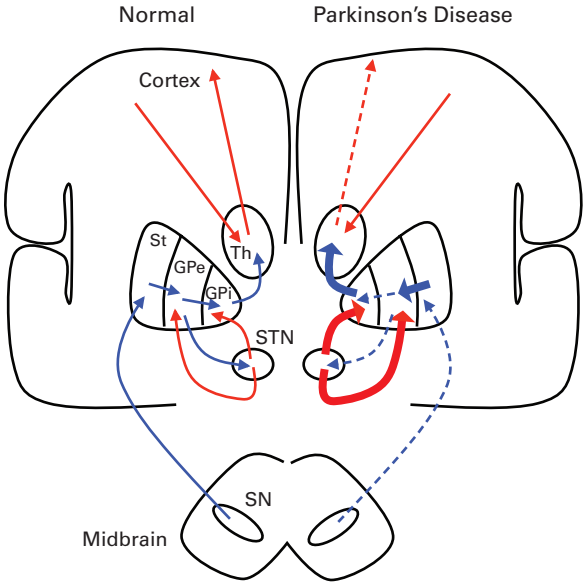


Plate 22
An extremely simplified schematic of network imbalance in Parkinson's disease. Excitation is shown in red, and inhibition in blue. The normal brain (left) is contrasted with the brain in the Parkinson's disease state on the right, where thickened (thinned) lines indicate an increase (decrease) in excitation (red) or inhibition (blue). St, striatum; GPe, globus pallidus externa; GPi, globus pallidus interna; Th, thalamus; STN, subthalamic nucleus; SN, substantia nigra. I have made no distinction between indirect and direct pathways, and customized this for the purposes of the discussion within this chapter. For a more complete and detailed description of this anatomy, see [OMRO+08].

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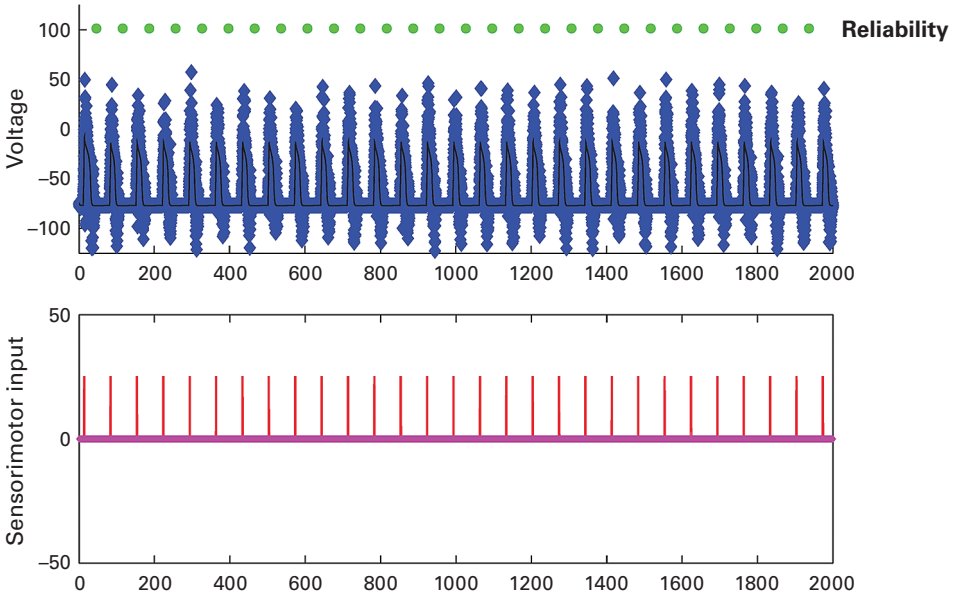


Plate 23
Normal response of TC cell to periodic sensorimotor stimulation. Voltage in reduced TC cell model (thin black line, upper panel), with substantial added noise to serve as a noisy observable (blue markers, upper panel), and sensorimotor input (red pulses, lower panel). Each time a sensorimotor pulse is reliably transmitted, a marker (green) is placed above the successfully transmitted spike. This cell is 100% reliable. Reproduced from [Sch10] with permission.

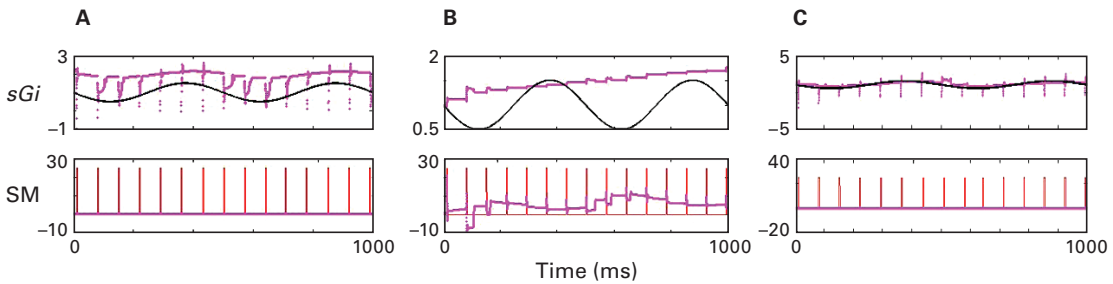


Plate 24
Tracking and estimating parameters as a function of process noises Q. Estimates of synaptic current from GPi (*sGi,* upper), and sensorimotor input (SM input, lower). Process noise parameters are (A) *sGi* Q = 30, *SM* Q= 0.01, (B) *sGi* Q = 0.01, *SM* Q = 30, and (C) *sGi* Q = 10.0, *SM* Q = 0.01. The algorithmic incorporation of such process noise can be explored in the code archive with [SS08]. Reproduced from [Sch10] with permission.

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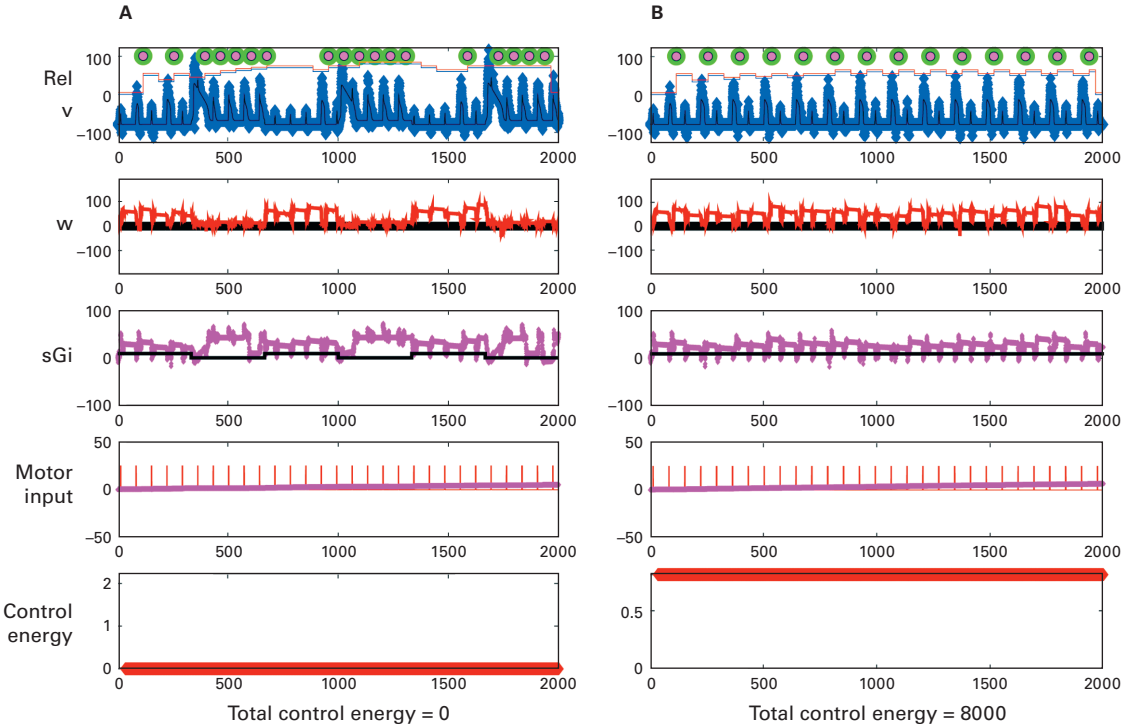


Plate 25
(A) Uncontrolled reduced TC cell dynamics in the Parkinsonian state with fluctuating current from GPi (sGi). (B) Perfect DBS stimulation filling in the troughs in the fluctuating current from GPi. Top panels show noisy observable voltage (blue symbols), reliability as piecewise continuous plots without (blue) and with (red) control on, the green circles are the timing of SM spikes, and the smaller red circles are transmitted spikes. The second panels show estimated w (red). The third panels show real (black) and estimated (magenta) synaptic current from GPi (sGi, estimated values multiplied by 10 for discriminability from the true values). The fourth panel shows real (red) and estimated (magenta) motor input (we are deliberately not trying to reconstruct motor input in the reconstruction through Q ratio adjustment). The bottom panel shows the running control energy, the squared value of the control signal at each time point, and the total sum of squares given as total control energy. Reproduced from [Sch10] with permission.

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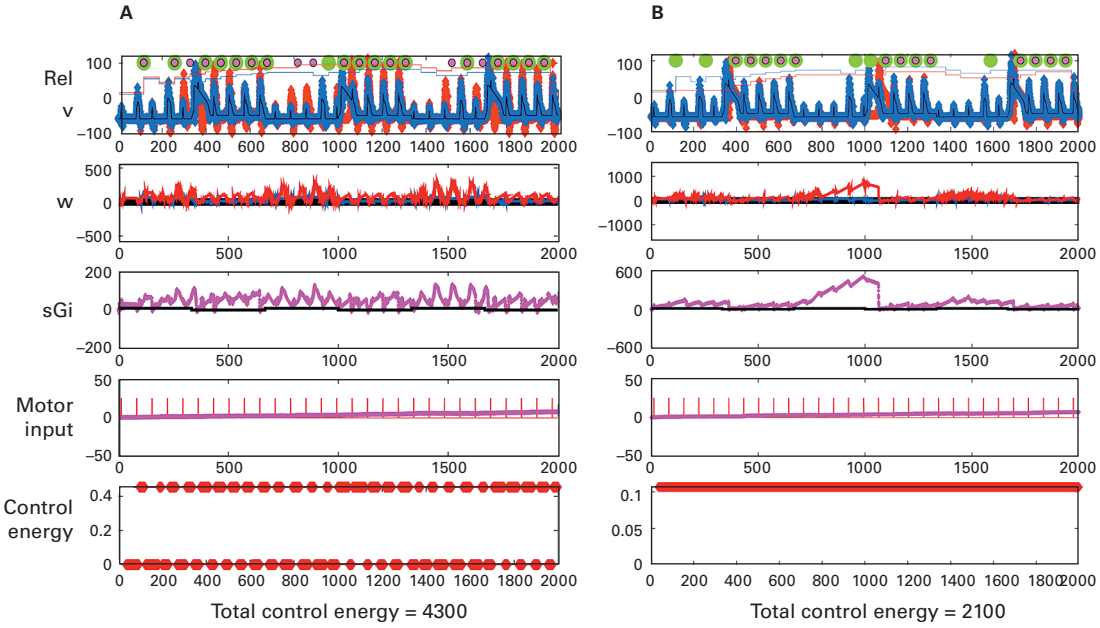


Plate 26
Feedback control scenario based on turning on and off deep brain stimulator based on a running average of the estimated T-current availability shown in A. B shows a scenario where a constant amount of deep brain stimulation is simply added to the fluctuating Parkinsonian GPi output. No adjustment of GPi constant stimulation comes close to the reliability achieved with the closed loop feedback scenario shown in A. Symbols as in figure 10.13. Reproduced from [Sch10] with permission.

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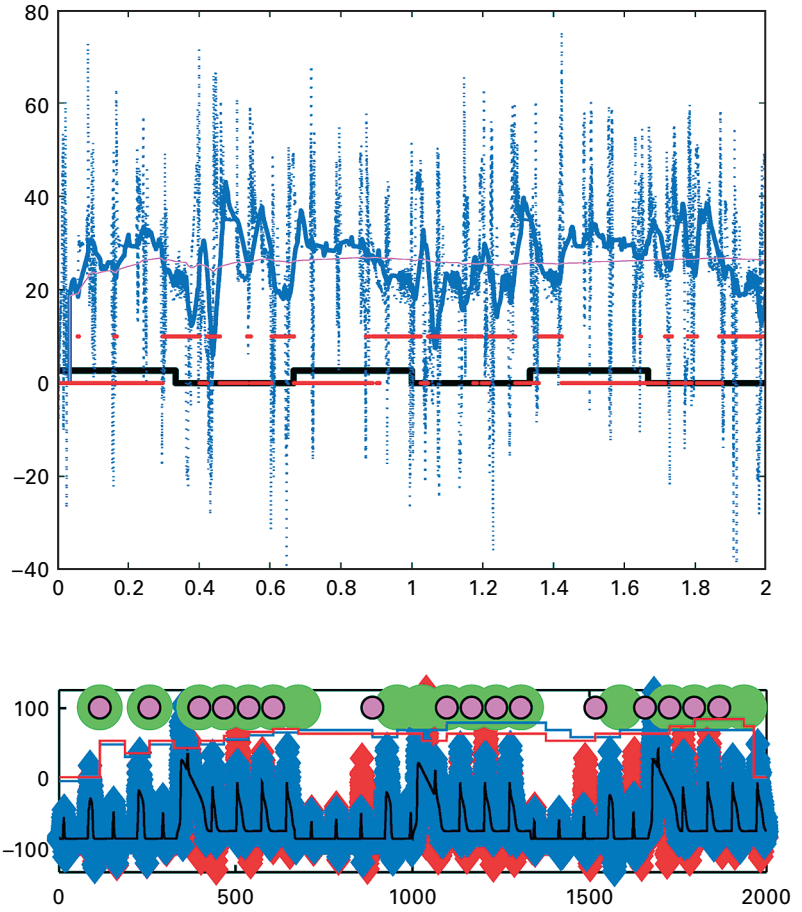


Plate 27
Estimated GPi input to TC cell (blue dotted line), and smoothed short-term moving average of this GPi input (blue solid line). We take a long-term moving average of this current (magenta line) as an adapting threshold to tell when the more instantaneous GPi input is fluctuating up or down. Crossing below the threshold determines when to turn the control on (red). The actual GPi fluctuations are shown (black lines). In the lower panel are the results with control off (blue markers), and on (red markers), and the uncontrolled (green markers) and controlled (magenta markers) spikes transmitted. The running reliability of the TC cell is plotted as a piecewise continuous line for uncontrolled (blue line) and controlled (red line) scenarios. Reproduced from [Sch10] with permission.

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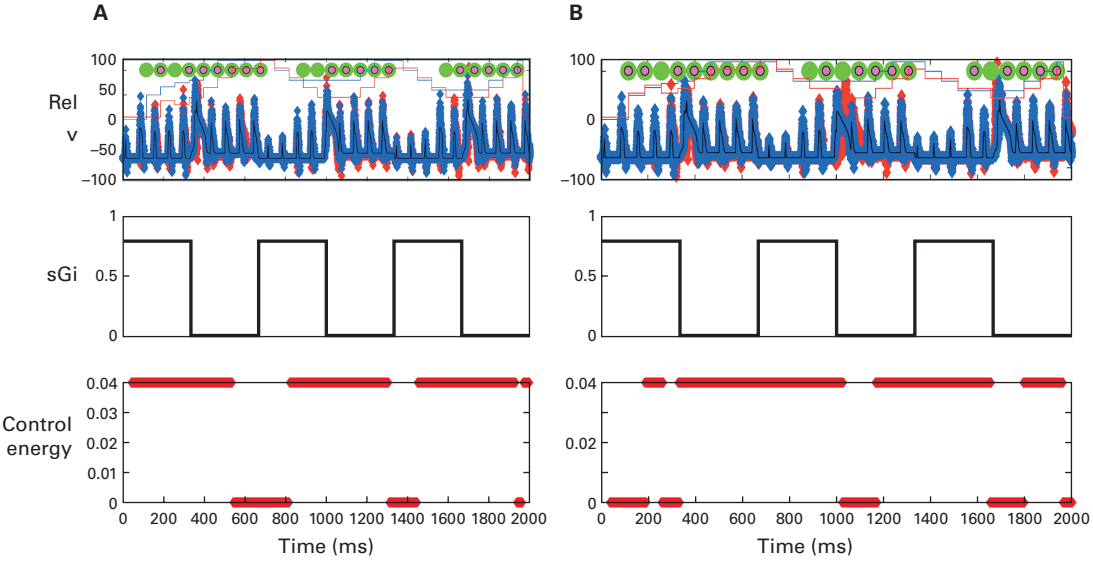


Plate 28
Control of a TC reduced-cell model using reliability as a control parameter. A shows a threshold of turning on GPi stimulation when reliability is <0.9. B shows a different strategy, using an inverse approach. In B, control is turned on when reliability is >0.5. Note that the relevant reliability in both examples is the controlled (red) piecewise continuous line in the upper panel (the blue reliability line is the uncontrolled state shown for comparison). The inherent delays in employing the moving average of reliability can be exploited so that inverse reliability control can be more reliable than when using a more intuitive strategy based on turning on stimulation when reliability falls. Reproduced from [Sch10] with permission.

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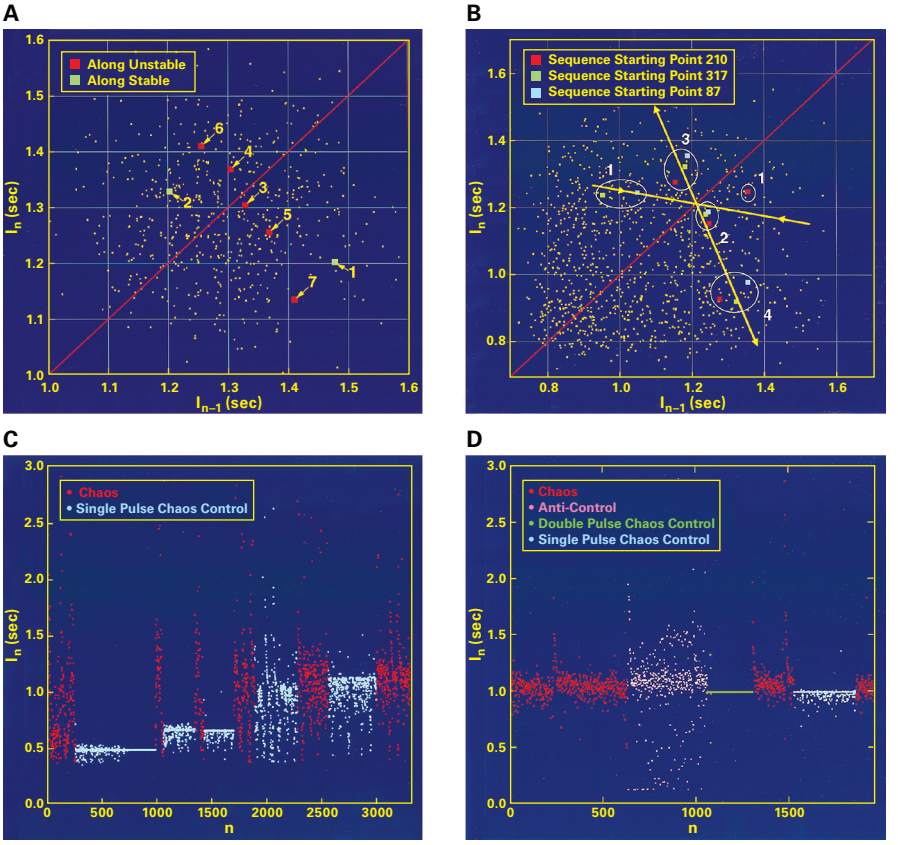


Plate 29

Controlling chaos in the brain. A shows the identification of a motif that shows an approach to the line of identity where sequential intervals, I_n and I_{n+1}, are equal. B shows multiple repetitions of this motif. C demonstrates that by directing the system to fall near the contracting direction (manifold), that the chaotic intervals could be regularized. D shows that by pushing the system away from the contracting direction, that we could prevent regularization. Reproduced from [SJD+94] with permission.

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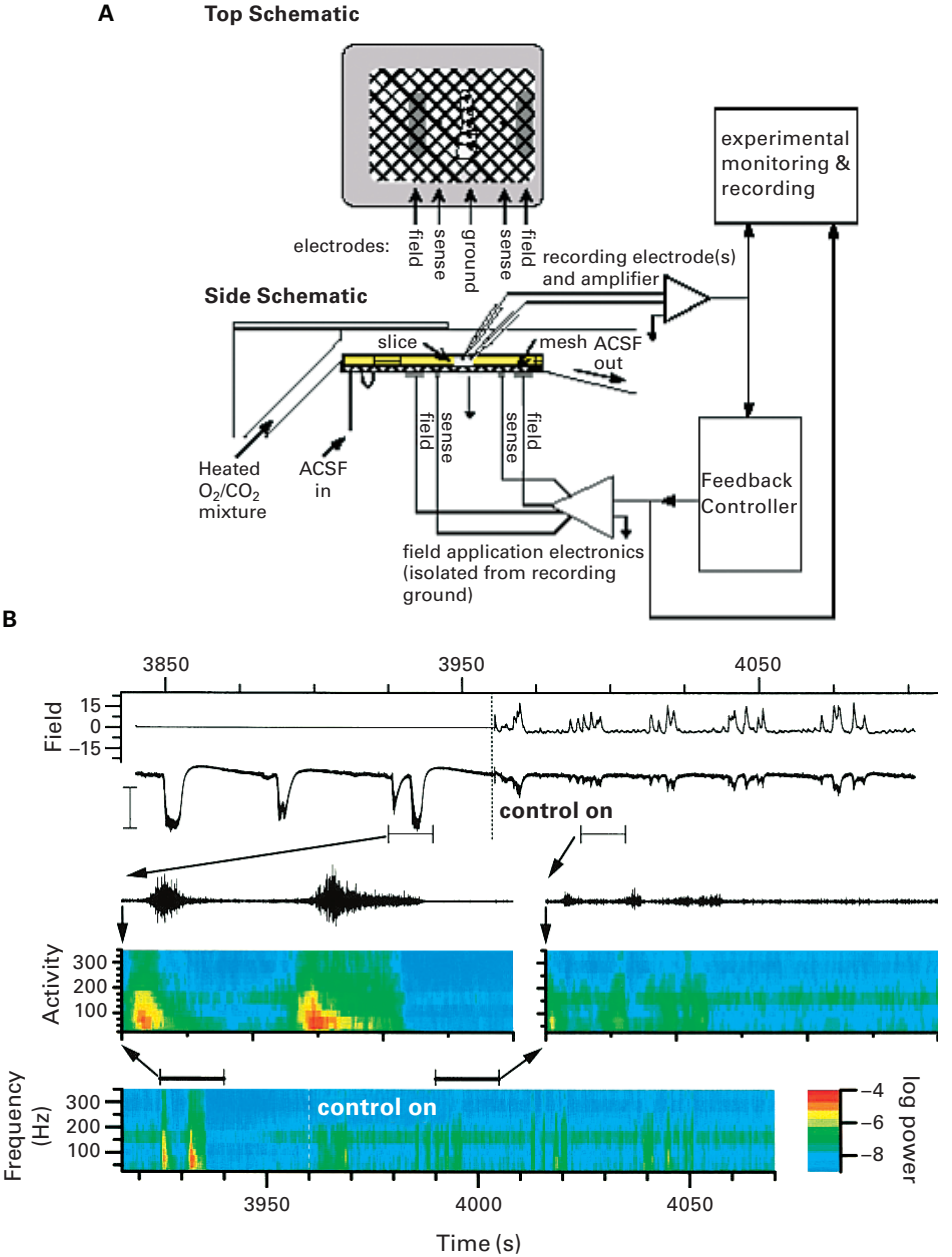


Plate 30
Adaptive feedback control of neuronal activity. Schematic of experimental chamber and electronics shown in A. In B are results demonstrating that a proportional control law could effectively suppress seizure activity in the brain slice. Reproduced from [GNWS01] with permission.

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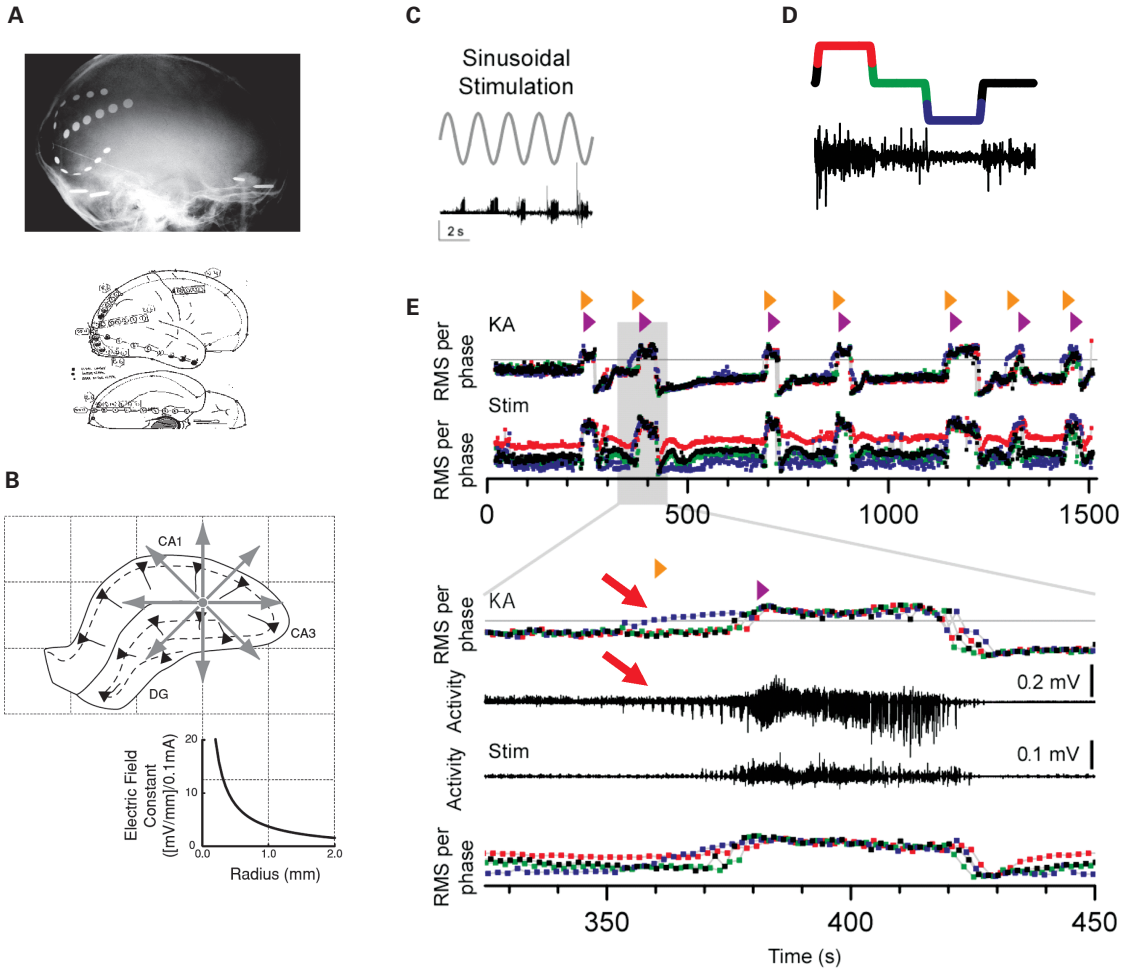


Plate 31

A shows an example of occipitally placed recording depth electrode in human epilepsy patient. Reproduced from [SSKW05] with permission. B is schematic of an axially placed stimulating electrode for epilepsy control in experimental hippocampal seizures. C shows experimental verification in animal models of sinusoidal entrainment of hippocampal neurons with low-frequency sinusoidal electrical fields. Using DC electrical fields of alternating polarity, in D, we demonstrated in E that by stimulating the contralateral side to the kainic acid (KA)–injected side with subthreshold alternating pulses, that the network demonstrated progressively increasing responses to these small stimuli in the seconds leading up to the start of the seizure (red arrows). B through E are reproduced from [RGW+03] with permission.

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